CHAPTER 2

Regression

2.1. Definition

Linear regression lives up to its name: it is a very straightforward approach for predicting a quantitative response Y on the basis of a predictor variable X. We assume that the true relationship between X and Y takes the form Y = f(X) + ε for some unknown function f, where ε is a mean-zero random error term. If f is to be approximated by a linear function, then we can write this relationship as:

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

Here β_0 is the intercept term—that is, the expected value of Y when X = 0, and β_1 is the slope—the average increase in Y associated with a one-unit increase in X. The error term: the true relationship is probably not linear, there may be other variables that cause variation in Y, and there may be measurement error. We typically assume that the error term is independent of X. The model given by this equation defines the *population regression line*, which is the best linear approximation to the true relationship between X and Y.

2.2. Linear regression model

There are two very different algorithms or approaches for implementing the Linear Regression model: the "closed-form" solution approach known as the Ordinary Least Squares (OLS) method and the iterative optimization approach known as the Gradient Descent (GD).

2.2.1. Closed-form solution: Ordinary Least Squares (OLS)

OLS is a type of linear least squares method for estimating the unknown parameters in a linear regression model. OLS chooses the parameters of a linear function of a set of explanatory variables by the principle of least squares:

minimizing the sum of the squares of the differences between the observed dependent variable (values of the variable being predicted) in the given dataset and those predicted by the linear function.

OLS finds the optimal parameters by computing a closed-form solution for the **Normal equation**.

The OLS method is suitable for the following scenario.

- Dataset: not too large and fits into computer's memory
- No. of Features: not too large
- Relationship between input (features) and output (target): Linear

a) Simple linear regression

Example: Let's consider the simple linear regression model in which the price of a house is related to the number of square feet of living area (SQFT).

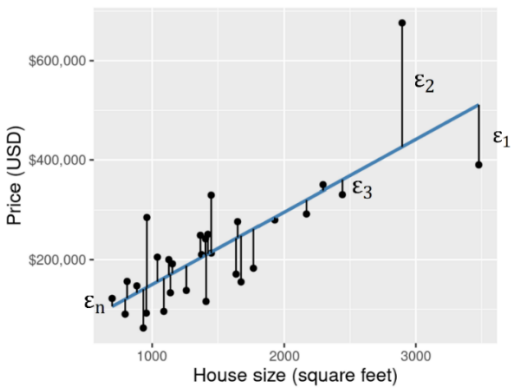


Figure 2.1 Scatter plot of sale price versus size

We have the system of equations:

$$\begin{cases} y_1 = \beta_0 + \beta_1 x_1 + \varepsilon_1 \\ y_2 = \beta_0 + \beta_1 x_2 + \varepsilon_2 \\ y_3 = \beta_0 + \beta_1 x_3 + \varepsilon_3 \\ \vdots \\ y_n = \beta_0 + \beta_1 x_n + \varepsilon_n \end{cases}$$

This system of equations can be written as:

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

So, we get:

$$\varepsilon = Y - X\beta$$

The L2 norm of ε is defined by:

$$\|\varepsilon\| = \sqrt{{\varepsilon_1}^2 + {\varepsilon_2}^2 + {\varepsilon_3}^2 + \cdots + {\varepsilon_n}^2} = \|Y - X\beta\|$$

The objective is to minimize the sum of the squared errors

$$\min_{\beta_0,\beta_1} \|\varepsilon\|^2 = \min_{\beta_0,\beta_1} \|Y - X\beta\|^2 = \min_{\beta_0,\beta_1} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 = \min_{\beta_0,\beta_1} E(\beta_0, \beta_1)$$

We have to find β_0 and β_1 such that:

$$\begin{cases} \dfrac{\partial E}{\partial \beta_0} = -2 \displaystyle\sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \dfrac{\partial E}{\partial \beta_1} = -2 \displaystyle\sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) x_i = 0 \end{cases}$$

$$\begin{cases} \displaystyle\sum_{i=1}^{n} y_i - n\beta_0 - \beta_1 \sum_{i}^{n} x_i = 0 \\ \displaystyle\sum_{i=1}^{n} y_i x_i - \beta_0 \sum_{i=1}^{n} x_i - \beta_1 \sum_{i=1}^{n} {x_i}^2 = 0 \end{cases}$$

$$\begin{cases} n\bar{y} - n\beta_0 - n\beta_1 \bar{x} = 0 \\ \displaystyle\sum_{i=1}^{n} y_i x_i - n\beta_0 \bar{x} - \beta_1 \sum_{i=1}^{n} {x_i}^2 = 0 \end{cases}$$

$$\begin{cases} \beta_0 = \bar{y} - \beta_1 \bar{x} \\ \displaystyle\sum_{i=1}^{n} y_i x_i - n\beta_0 \bar{x} - \beta_1 \sum_{i=1}^{n} {x_i}^2 = 0 \end{cases}$$

We plug the first equation in the second one, and we get:

$$\sum_{i=1}^{n} y_i x_i - n\bar{x}(\bar{y} - \beta_1 \bar{x}) - \beta_1 \sum_{i=1}^{n} {x_i}^2 = 0$$

$$\sum_{i=1}^{n} y_i x_i - n\bar{x}\bar{y} + \beta_1 n\bar{x}^2 - \beta_1 \sum_{i=1}^{n} {x_i}^2 = 0$$

$$\sum_{i=1}^{n} y_i x_i - n\bar{x}\bar{y} = \beta_1 (\sum_{i=1}^{n} {x_i}^2 - n\bar{x}^2)$$

$$\beta_1 = \frac{\sum_{i=1}^{n} y_i x_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n} {x_i}^2 - n\bar{x}^2}$$

After simplification, we get:

$$\beta_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\beta_1 = \frac{cov(x,y)}{var(x)}$$

b) Multiple linear regression model

Example: Suppose that Y is the fuel consumption of a particular model of car in m.p.g.

Suppose that the attributes are:

1. X1 — the weight of the car.
2. X2 — the horse power.
3. X3 — the no. of cylinders.

Typically, the data will be available in the form of an array like this:

$$\begin{array}{cccc} y_1 & x_{11} & x_{12} & x_{13} \\ y_2 & x_{21} & x_{22} & x_{23} \\ \dots & & \dots & \\ y_n & x_{n1} & x_{n2} & x_{n3} \end{array}$$

where n is the number of observations or cases in the dataset.

One very general form for the model would be:

$$Y = f(X_1, X_2, X_3) + \varepsilon$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

Given the actual data, we may write

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i \quad i = 1, \dots, n$$

But the use of subscripts becomes inconvenient and conceptually obscure. We will find it simpler both notationally and theoretically to use a matrix/vector representation. The regression equation is written:

$$y = X\beta + \varepsilon$$

where $y = (y_1, \dots, y_n)^T$, $\beta = (\beta_0, \dots, \beta_3)^T$ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^T$

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ \dots & & \dots & \\ 1 & x_{n1} & x_{n2} & x_{n3} \end{pmatrix}$$

The column of ones incorporates the intercept term.

Estimating β

The problem is to find β such that Xβ is close to Y. We might define the best estimate of β as that which minimizes the sum of the squared errors, $\varepsilon^T \varepsilon$. That is to say that the least squares estimate of β, called $\hat{\beta}$ minimizes:

$$\|\varepsilon\|_2^2 = \varepsilon^T \varepsilon = (y - X\beta)^T (y - X\beta)$$

Note: Basic properties of transpose:

1. Double transpose:
$$(A^T)^T = A$$
2. Transpose of a sum:
$$(A + B)^T = A^T + B^T$$
3. Transpose of a scalar multiple:
$$(cA)^T = cA^T$$
4. Transpose of a product:
$$(AB)^T = B^T A^T$$

By applying the transpose operations, we get:

$$(y^T - (X\beta)^T)(y - X\beta)$$
$$y^T y - y^T X\beta - (X\beta)^T y + (X\beta)^T (X\beta)$$
$$y^T y - y^T X\beta - \beta^T X^T y + \beta^T X^T X\beta$$

$y^T X\beta$ is a scalar (a 1x1 matrix), so we apply the symmetry rule: $a^T b = b^T a$

$$y^T y - (X\beta)^T y - \beta^T X^T y + \beta^T X^T X\beta$$
$$y^T y - \beta^T X^T y - \beta^T X^T y + \beta^T X^T X\beta$$
$$y^T y - 2\beta^T X^T y + \beta^T X^T X\beta$$

Differentiating with respect to β and setting to zero, we find that $\hat{\beta}_{OLS}$ satisfies

$$-2X^T y + 2X^T X\hat{\beta}_{OLS} = 0$$
$$X^T X\hat{\beta}_{OLS} = X^T y$$

Provided $X^T X$ is invertible, the OLS solution is:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

$X^T X$ is called a Gram matrix, and captures relationships between features.

If collinearity between features is high, then $X^\top X$ is almost singular (i.e., it's not full rank and its determinant is close to zero). It's almost impossible to invert it accurately. Because the OLS estimates depend upon $(X^T X)^{-1}$, we would have problems in computing $\hat{\beta}_{OLS}$.

If $X^\top X$ is nearly singular, $(X^T X)^{-1}$ has very large values. In such cases, small changes to the elements of X can lead to large variations in $(X^T X)^{-1}$, resulting in inflated coefficients in $\hat{\beta}_{OLS}$. As a result, the OLS estimates become unstable and highly sensitive to small changes in the data.

The variance of the estimated coefficients in Ordinary Least Squares (OLS) is given by:

$$Var(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

- This makes $(X^T X)^{-1}$ very large, which increases the variance of $\hat{\beta}$. Different samples from the data can produce very different estimates of $\hat{\beta}$. The least square estimator $\hat{\beta}$ may provide a good fit to the training data, but it will not fit sufficiently well to the test data.

2.3. Regression shrinkage (regularization) methods

2.3.1. Ridge Regression

A possible way to address this issue is to relax the constraint of having an unbiased estimator. The goal then becomes minimizing the sum of squared errors:

$$\min_{\beta}\|\varepsilon\|^2 = \min_{\beta}\|Y - X\beta\|^2, s.t. \|\beta\|^2 \leq c^2$$

Geometric Interpretation of Ridge Regression:

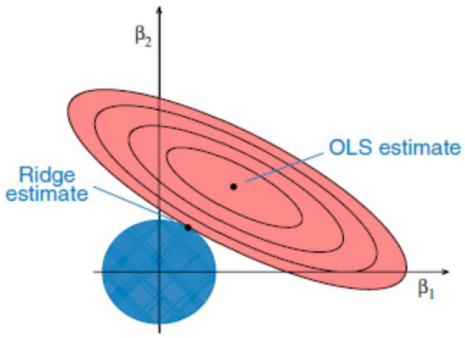


Figure 2.2 Illustration of ridge regression estimation. The plot displays the contours of both the error function and the constraint. The solid blue region represents the constraint set $\beta_1^2 + \beta_2^2 \leq c^2$, whereas the red ellipses correspond to the contours of the least-squares error. Ridge regression reduces the magnitude of all coefficients without setting any of them exactly to zero; the resulting coefficients are smaller but remain nonzero.

The ellipses correspond to the contours of the sum of the squared errors: the inner ellipse has smaller sum of the squared errors, and the sum of the squared errors is minimized at ordinal least square (OLS) estimates.

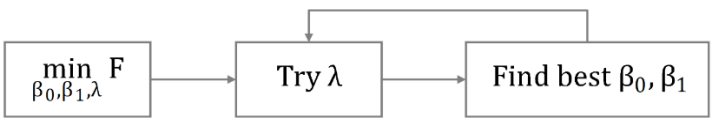
For p=2, the constraint in ridge regression corresponds to a circle: $\sum_{j=1}^{p}\beta_j^2 \leq c^2$

We want to minimize:

$$\min_{\beta}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2, s.t. \beta_0^2 + \beta_1^2 \leq c^2$$

Instead of solving this as a constrained optimization problem, it is typically rewritten using a Lagrange multiplier technique and solved as an unconstrained optimization problem:

$$\min_{\beta} F(\beta_0, \beta_1, \lambda) = \min_{\beta}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2 + \lambda(\beta_0^2 + \beta_1^2 - c^2)$$



So, we have to minimize:

$$\min_{\beta}\|Y - X\beta\|^2 + \lambda\|\beta\|^2$$

The ridge solution:

$$\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

2.3.2. Lasso Regression

A ridge solution can be hard to interpret because it is not sparse (no β's are set exactly to 0). What if we constrain the L1 norm instead of the Euclidean (L2) norm?

The objective is to minimize the sum of the squared errors:

$$\min_{\beta}\|\varepsilon\|^2 = \min_{\beta}\|Y - X\beta\|^2, s.t. \|\beta\|_1 \leq c$$

Geometric Interpretation of Lasso Regression:
The lasso performs L1 shrinkage, so that there are "corners" in the constraint, which in two dimensions corresponds to a diamond. If the sum of squares "hits" one of these corners, then the coefficient corresponding to the axis is shrunk to zero.

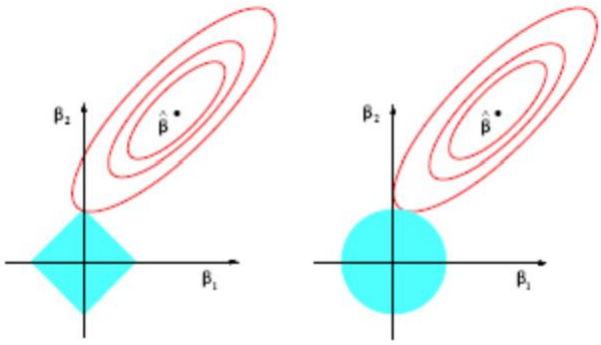


Figure 2.3 Estimation picture of the Lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions $|\beta_1| + |\beta_2| \leq c$ and $\beta_1^2 + \beta_2^2 \leq c^2$, respectively, while the red ellipses are the contours of the least squares error function. Lasso shrinks some coefficients to zero, resulting in sparse models.

For p=2, the constraint in lasso regression corresponds to a diamond-shaped region: $\sum_{j=1}^{p}|\beta_j| \leq c$

We want to minimize:

$$\min_{\beta}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2, s.t. |\beta_0| + |\beta_1| \leq c$$

As p increases, the multidimensional diamond has an increasing number of corners, and so it is highly likely that some coefficients will be set equal to zero. Hence, the lasso performs shrinkage and (effectively) subset selection.
Instead of solving the constrained form, we rewrite it using Lagrange multiplier technique:

$$\min_{\beta}\|Y - X\beta\|^2 + \lambda\|\beta\|_1$$

Unlike ridge regression, Lasso does not have a simple closed-form solution because the **L1-norm is not differentiable** at zero. Instead, numerical optimization techniques are used, like gradient descent.

2.3.3. Elastic Net Regression

Elastic Net regression combines the strengths of Ridge and Lasso regression, which involves both penalties $L1$ and $L2$. For Elastic Net regression, the objective function is:

$$\min_{\beta}\|Y - X\beta\|^2, s.t. (1-\rho)\|\beta\|^2 + \rho\|\beta\|_1 \leq c$$

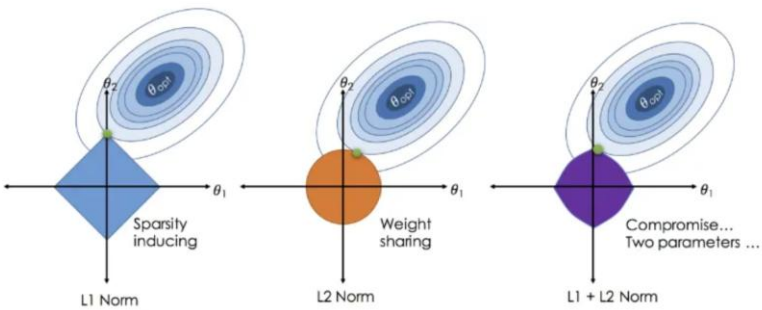


Figure 2.4 Estimation picture of the Lasso (left), ridge regression (middle), and elastic net (right). Shown are contours of the error and constraint functions. In the right, the solid purple area is the constraint region $(1-\rho)\|\beta\|^2 + \rho\|\beta\|_1 \leq c$. Elastic Net shrinks coefficients like Lasso but distributes the penalty across correlated predictors, often retaining groups of features.

Instead of solving the constrained form, we rewrite it using a Lagrange multiplier technique:

$$\min_{\beta} \|Y - X\beta\|^2 + \lambda((1-\rho)\|\beta\|^2 + \rho\|\beta\|_1)$$

Ridge vs. Lasso vs. Elastic Net

Method	Effect on Weak Features	Effect on Collinear Features	Feature Selection?
Lasso (L1)	Eliminates them (sets to zero)	Picks one and removes others	Yes
Ridge (L2)	Shrinks them (but keeps all)	Spreads importance among them	No
Elastic Net	Eliminates weak ones	Keeps multiple collinear ones	Yes (but smoother than Lasso)

Conclusion

Why might we want to use another fitting procedure instead of least squares? As we will see, alternative fitting procedures can yield better prediction accuracy and model interpretability.

- *Prediction Accuracy:* Provided that the true relationship between the response and the predictors is approximately linear, the least squares estimates will have low bias. If $n \gg p$ — that is, if n, the number of observations, is much larger than p, the number of variables — then the least squares estimates tend to also have low variance, and hence will perform well on test observations. However, if n is not much larger than p, then there can be a lot of variability in the least squares fit, resulting in overfitting and consequently poor predictions on future observations not used in model training. And if $p > n$, then there is no longer a unique least squares coefficient estimate: there are infinitely many solutions. Each of these least squares solutions gives zero error on the training data, but typically very poor test set performance due to extremely high variance. By constraining or shrinking the estimated coefficients, we can often substantially reduce the variance at the cost of a negligible increase in bias. This can lead to substantial improvements in the accuracy with which we can predict the response for observations not used in model training.
- *Model Interpretability:* It is often the case that some or many of the variables used in a multiple regression model are in fact not associated with the response. Including such irrelevant variables

leads to unnecessary complexity in the resulting model. By removing these variables — that is, by setting the corresponding coefficient estimates to zero — we can obtain a model that is more easily interpreted. Now least squares is extremely unlikely to yield any coefficient estimates that are exactly zero. Some approaches for automatically performing feature selection or variable selection can be used for excluding irrelevant variables from a multiple regression model.

There are many alternatives, both classical and modern, to using least squares. The three important classes of methods are:

1) **Shrinkage:** This approach involves fitting a model involving all p predictors. However, the estimated coefficients are shrunken towards zero relative to the least squares estimates. This shrinkage (also known as regularization) has the effect of reducing variance. Depending on what type of shrinkage is performed, some of the coefficients may be estimated to be exactly zero. Hence, shrinkage methods can also perform variable selection.
2) **Subset Selection:** This approach involves identifying a subset of the p predictors that we believe to be related to the response. We then ft a model using least squares on the reduced set of variables.
3) **Dimension Reduction:** This approach involves projecting the p predictors into an M-dimensional subspace, where M different linear combinations, or projections, of the variables. Then these M projections are used as predictors to ft a linear regression model by least squares.

2.4. Goodness of Fit

How well does the model fit the data? One measure is R^2, the so-called coefficient of determination or percentage of variance explained

$$R^2 = 1 - \frac{SSR}{SST}$$

Residual Sum of Squares (SSR): Unexplained variation (error)

$$SSR = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Total Sum of Squares (SST): Measures the total variation in the dependent variable (y) around its mean ($\bar{y}$)

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

$R^2 = 1$: model explains all variability (perfect fit)
$R^2 = 0$: model explains none of the variability (just predicts the mean)
$0 < R^2 < 1$: fraction of variance explained by the model

2.5. Polynomial Regression

If we have nonlinear relationship between the input variables (features) and the output variable (target), then we may create new features by adding powers of each feature to fit the nonlinear line. Finally, we train a linear model on this augmented set of features. This technique is called Polynomial Regression. The degree of a Polynomial Regression determines the complexity of the model.

The standard way to extend linear regression to settings in which the relationship between the predictors and the response is nonlinear has been to replace the standard linear model.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

with a polynomial function

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \cdots + \beta_d x_i^d + \epsilon_i,$$

where ε_i is the error term.
For large enough degree d, a polynomial regression allows us to produce an extremely non-linear curve. Notice that the coefficients can be easily estimated using least squares linear regression because this is just a standard linear model with predictors x_i, x_i^2, x_i^3, …, x_i^d. Generally speaking, it is unusual to use d greater than 3 or 4 because for large values of d, the polynomial curve can become overly flexible and can take on some very strange shapes.

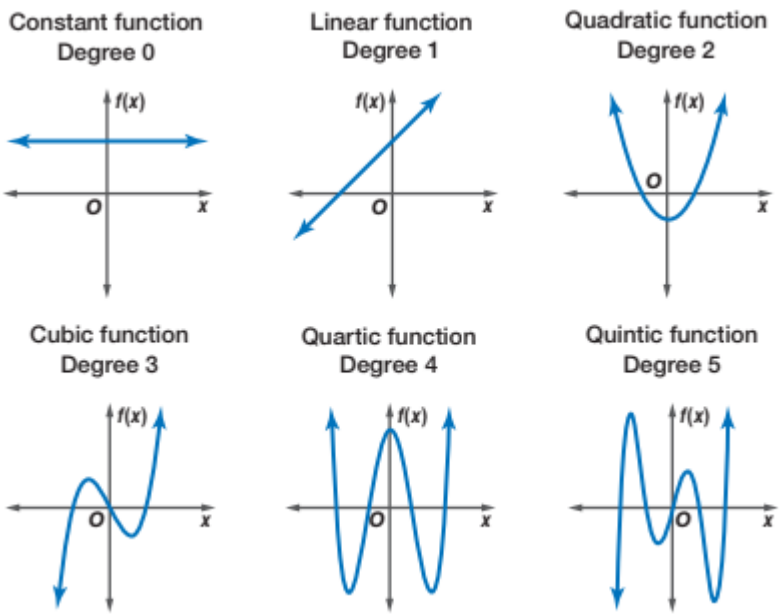


Figure 2.5 Polynomial curves

Matrix form and calculation of estimates

We have the system of equations:

$$\begin{cases} y_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_1^3 + \cdots + \beta_d x_1^d + \varepsilon_1 \\ y_2 = \beta_0 + \beta_1 x_2 + \beta_2 x_2^2 + \beta_3 x_2^3 + \cdots + \beta_d x_2^d + \varepsilon_2 \\ y_3 = \beta_0 + \beta_1 x_3 + \beta_2 x_3^2 + \beta_3 x_3^3 + \cdots + \beta_d x_3^d + \varepsilon_3 \\ \vdots \\ y_n = \beta_0 + \beta_1 x_n + \beta_2 x_n^2 + \beta_3 x_n^3 + \cdots + \beta_d x_n^d + \varepsilon_n \end{cases}$$

This system of equations can be written as:

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 & x_1^3 & \dots & x_1^d \\ 1 & x_2 & x_2^2 & x_2^3 & \dots & x_2^d \\ 1 & x_3 & x_3^2 & x_3^3 & \dots & x_3^d \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 & \dots & x_n^d \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_d \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

which when using pure matrix notation is written as:

$$y = X\beta + \varepsilon$$

The vector of estimated polynomial regression coefficients (using OLS) is:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

General case

In polynomial regression with multiple features, we extend the formulation to include higher-order terms and cross-terms. Let's assume we have *m* features $x_1, x_2, ..., x_m$ and want to fit a polynomial of degree d. The general form of the model is:

$$y = \beta_0 + \sum_{j=1}^{m} \beta_j x_j + \sum_{j=1}^{m} \sum_{k=j}^{m} \beta_{jk} x_j x_k + \cdots + \beta_d \prod_{j=1}^{m} x_j^{d_j} + \epsilon$$

the terms include all polynomial combinations of the features up to degree d.

Example 1: for 2 Features x_1, x_2 and d=2:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \epsilon$$

The design matrix X is

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{11}^2 & x_{12}^2 & x_{11}x_{12} \\ 1 & x_{21} & x_{22} & x_{21}^2 & x_{22}^2 & x_{21}x_{22} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n1}^2 & x_{n2}^2 & x_{n1}x_{n2} \end{bmatrix}$$

The coefficient vector

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{bmatrix}$$

The output vector

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Example 2: For a third-degree polynomial regression with three features (x_1, x_2, x_3), the model includes all possible monomials up to degree 3:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1^2 + \beta_5 x_2^2 + \beta_6 x_3^2 + \beta_7 x_1 x_2 + \beta_8 x_1 x_3 + \beta_9 x_2 x_3$$

$$+\beta_{10} x_1^3 + \beta_{11} x_2^3 + \beta_{12} x_3^3 + \beta_{13} x_1^2 x_2 + \beta_{14} x_1^2 x_3 + \beta_{15} x_2^2 x_1 + \beta_{16} x_2^2 x_3 + \beta_{17} x_3^2 x_1 + \beta_{18} x_3^2 x_2 + \beta_{19} x_1 x_2 x_3 + \epsilon$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & x_{11}^2 & x_{12}^2 & x_{13}^2 & x_{11}x_{12} & x_{11}x_{13} & x_{12}x_{13} & x_{11}^3 & x_{12}^3 & x_{13}^3 & x_{11}^2x_{12} & x_{11}^2x_{13} & x_{12}^2x_{11} & x_{12}^2x_{13} & x_{13}^2x_{11} & x_{13}^2x_{12} & x_{11}x_{12}x_{13} \\ 1 & x_{21} & x_{22} & x_{23} & x_{21}^2 & x_{22}^2 & x_{23}^2 & x_{21}x_{22} & x_{21}x_{23} & x_{22}x_{23} & x_{21}^3 & x_{22}^3 & x_{23}^3 & x_{21}^2x_{22} & x_{21}^2x_{23} & x_{22}^2x_{21} & x_{22}^2x_{23} & x_{23}^2x_{21} & x_{23}^2x_{22} & x_{21}x_{22}x_{23} \\ 1 & x_{31} & x_{32} & x_{33} & x_{31}^2 & x_{32}^2 & x_{33}^2 & x_{31}x_{32} & x_{31}x_{33} & x_{32}x_{33} & x_{31}^3 & x_{32}^3 & x_{33}^3 & x_{31}^2x_{32} & x_{31}^2x_{33} & x_{32}^2x_{31} & x_{32}^2x_{33} & x_{33}^2x_{31} & x_{33}^2x_{32} & x_{31}x_{32}x_{33} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n3} & x_{n1}^2 & x_{n2}^2 & x_{n3}^2 & x_{n1}x_{n2} & x_{n1}x_{n3} & x_{n2}x_{n3} & x_{n1}^3 & x_{n2}^3 & x_{n3}^3 & x_{n1}^2x_{n2} & x_{n1}^2x_{n3} & x_{n2}^2x_{1} & x_{n2}^2x_{n3} & x_{n3}^2x_{n1} & x_{n3}^2x_{n2} & x_{n1}x_{n2}x_{n3} \end{bmatrix}$$

The coefficient vector is:

$$\boldsymbol{\beta}^T = \begin{bmatrix}\beta_0 & \beta_1 & \beta_2 & \beta_3 & \beta_4 & \beta_5 & \beta_6 & \beta_7 & \beta_8 & \beta_9 & \beta_{10} & \beta_{11} & \beta_{12} & \beta_{13} & \beta_{14} & \beta_{15} & \beta_{16} & \beta_{17} & \beta_{18} & \beta_{19}\end{bmatrix}^T$$

The number of polynomial terms:

$$p = \binom{m+d}{d} = \binom{3+3}{3} = \binom{6}{3} = \frac{6!}{3!\,(6-3)!} = 20$$